

Acton Research Technical Support SpectraPro Diagnostic Procedure



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Introduction to the Acton Research Technical Support

SpectraPro Diagnostic Procedure

This document describes the diagnostic steps taken by Acton Research Corporation (ARC) Technical support staff when you telephone ARC regarding a problem with your SpectraPro system. While you are on the telephone, the support staff will ask you to provide information about your computer system and may request remote access (via HyperTerminal) to the equipment for troubleshooting purposes.

The flowchart shows the major steps in the process. The details are presented in the pages that follow. Wherever possible screen shots (Windows® 2000-based) have been provided to illustrate the steps. Depending on your Windows® version, the dialogs that you see may not be identical to those in this document.

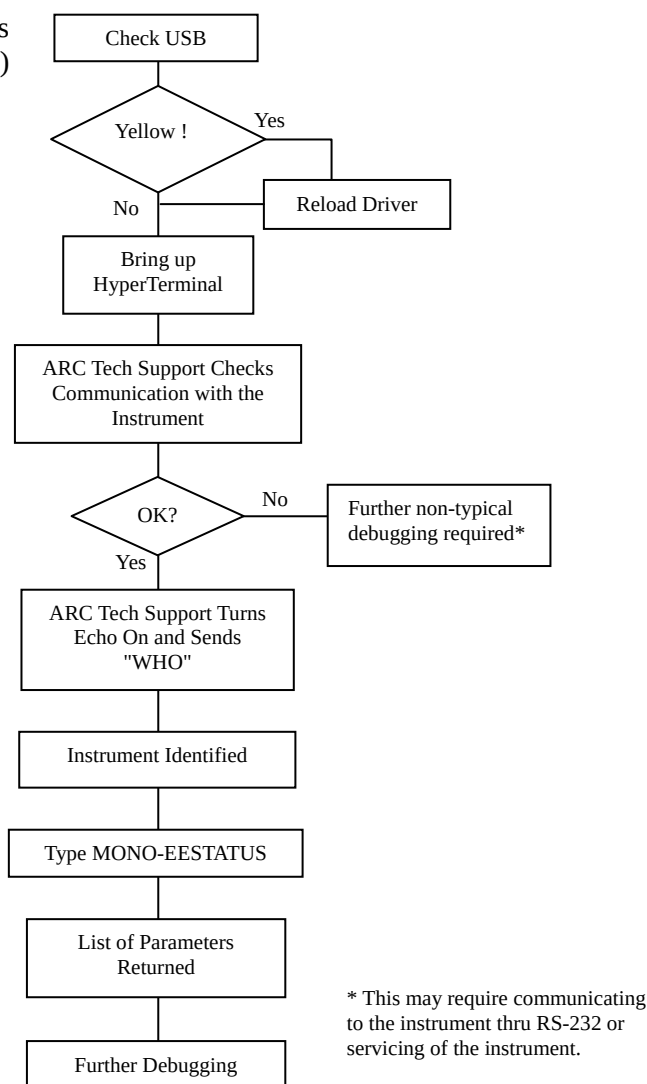


Figure 1. Diagnostic Flowchart



Checking the USB Drivers

Here are the initial steps that ARC goes through when diagnosing a SpectraPro through USB.

Check the USB drivers: This means that you will need to access System information via the Windows Control Panel. See Figure 2.

1. Boot your computer and click on the Windows 'Start' button
2. Click on 'Settings'
3. Click on 'Control Panel'
4. Click on 'System' icon

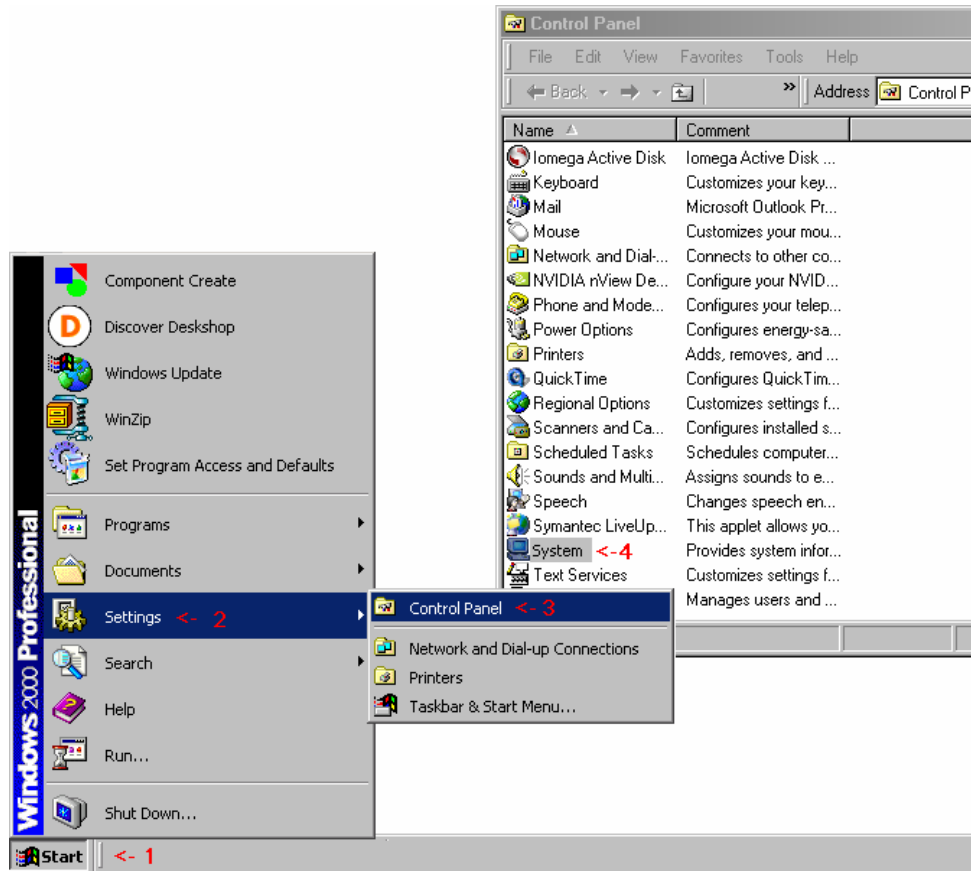


Figure 2. Open the Control Panel



5. Click on the 'Device Manager'
6. The USB driver will appear under Universal Serial Bus controllers
7. The Serial port will appear under Ports. Note the COM number associated with it (it is COM42 in the example below). We will need this number later.

If there is a **Yellow !** exclamation point on either device, the Acton Research USB driver needs to be reloaded. Go directly to page 7 for Reloading instructions. The directions there assume that you have opened the Device Manager.

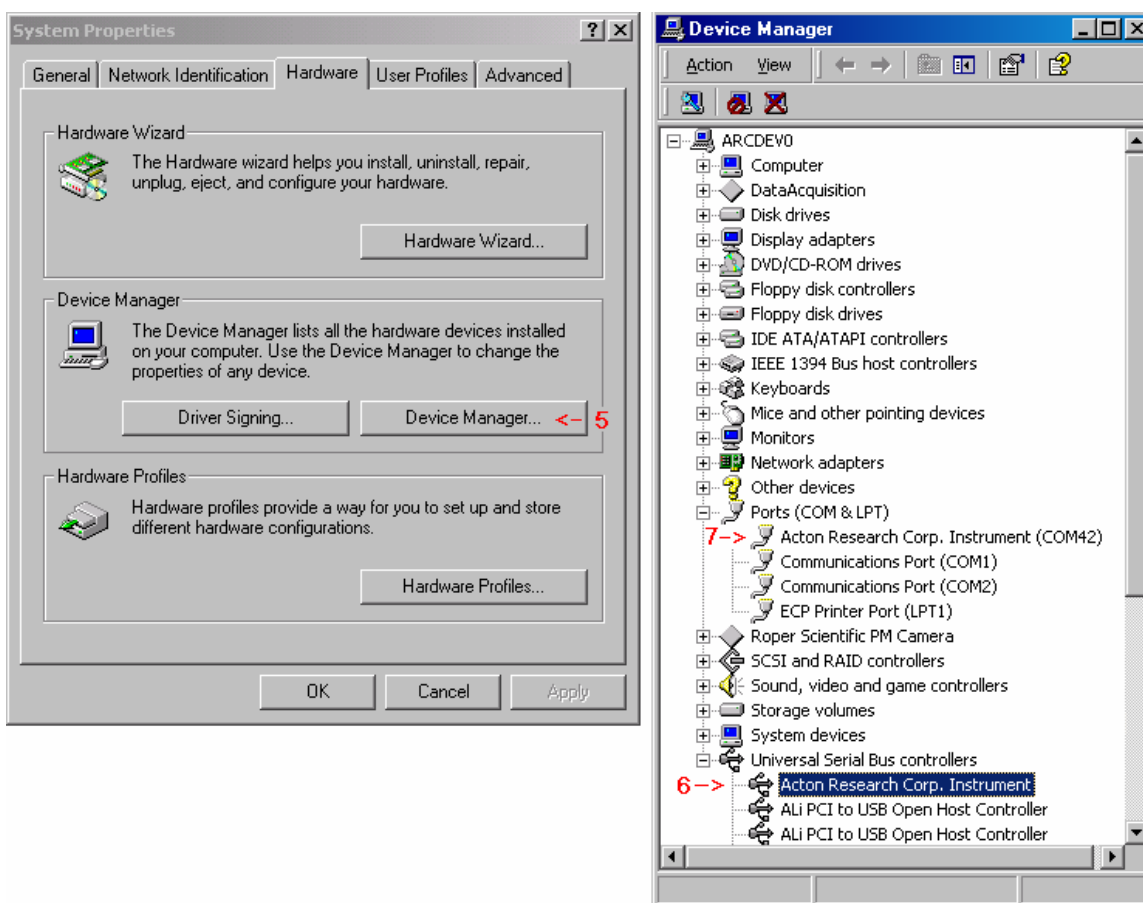


Figure 3. Opening the Device Manager





Opening HyperTerminal

You have checked the status of the device drivers and if required have reloaded them (see page 7 for Reloading instructions). You will now be asked to bring up HyperTerminal to talk to the instrument. (Commands can be found in the manual.) HyperTerminal, which comes with Windows 95, 98, Me, NT, 2K & XP, can be used to obtain diagnostic information from your instrument.

1. On the Windows Start panel, click on 'Programs'
2. Click on 'Accessories'
3. Click on 'Communications'
4. Click on 'HyperTerminal'

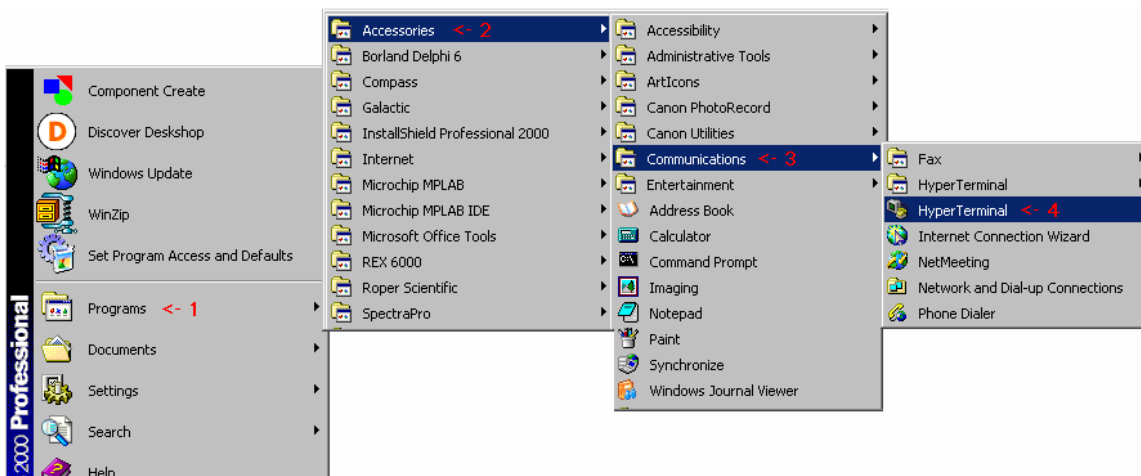


Figure 4. Open the HyperTerminal Program



5. Enter a name and choose an icon.
6. Click on 'OK' to open the Connect To dialog.
7. Select the COM port (as noted on page 5, step 7) from the 'Connect using' list (it is 'COM42' in the example below.)

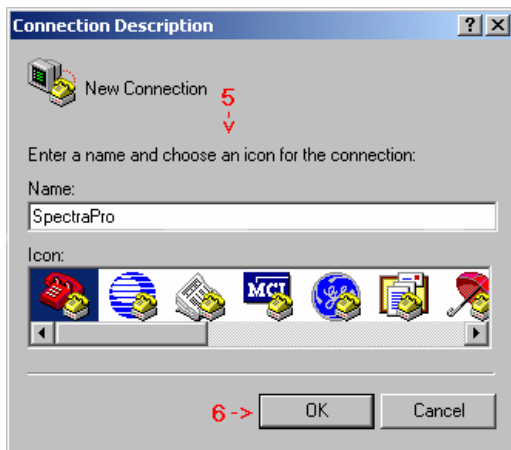


Figure 5. Entering a Name and Choosing an Icon

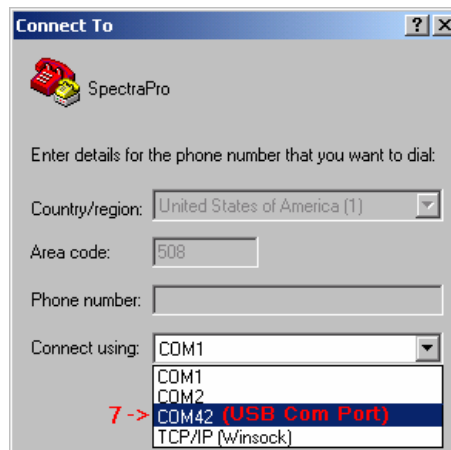


Figure 6. Selecting the COM Port

8. Select **9600** from the 'Bits per second' list, select **None** from the 'Flow control' list and then click on 'OK'.

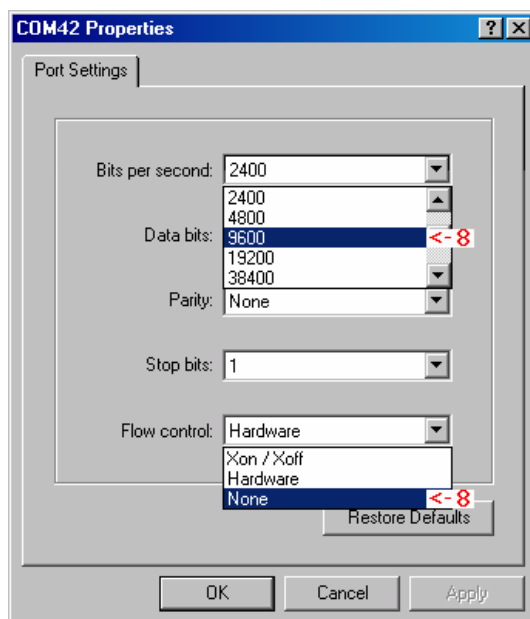


Figure 7. Selecting the Bits per Second (Speed) and Flow Control





9. Once HyperTerminal has been brought up we will see if we can talk to the instrument by hitting the 'Enter' key several times. Each strike should return an 'ok'. If we are communicating through USB, we will next turn on the terminal echo with 'echo-on' followed by the enter key. We will then send a 'WHO' to the instrument and request the result. As a final step we will have you type 'MONO-EESTATUS' and request the result. This request will return a long list of parameters that we will need to see to further debug the issue with the instrument.

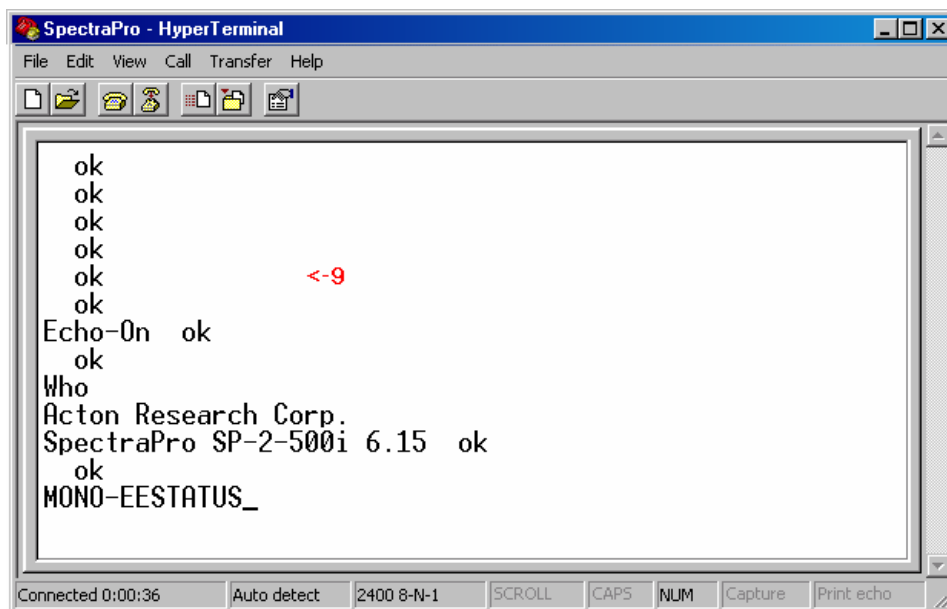


Figure 8. HyperTerminal screen





Reloading Drivers

These directions assume that you have opened the Device Manager.

If a **Yellow !** exclamation point appeared when you checked the USB drivers (see pages 2-3), you will need to reinstall the driver(s). The following text and screen captures step you through the process.

1. Under 'Universal Serial Bus controllers', double-click on **Acton Research Corp. Instrument** to open the Acton Research Corp. Instrument Properties dialog. Then, click on the 'Driver' tab.
2. Click on 'Update Driver'.
3. Click on the 'Next' button.

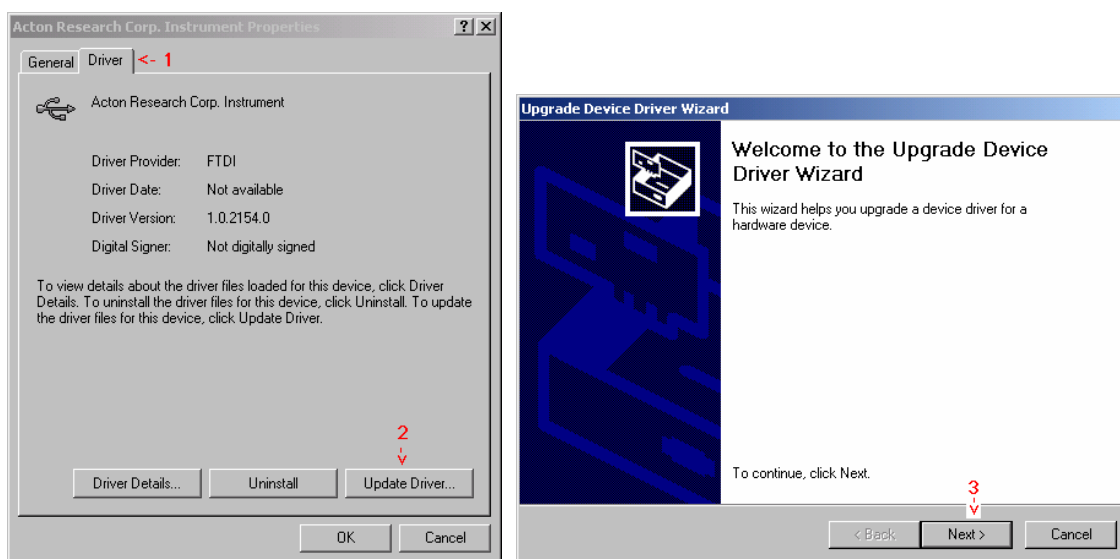


Figure 9. Opening the Driver Properties dialog





4. Select the 'Search for a suitable driver ...' radio button and click on 'Next'.

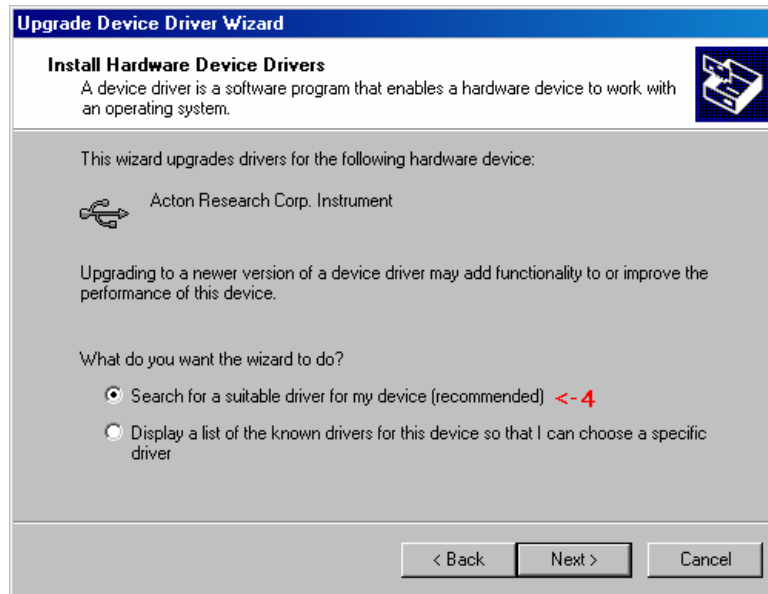


Figure 10. Install Hardware Device Drivers dialog

5. Put the Acton Research Corporation CD (supplied with your system) in the CDROM drive and click on 'Next'.

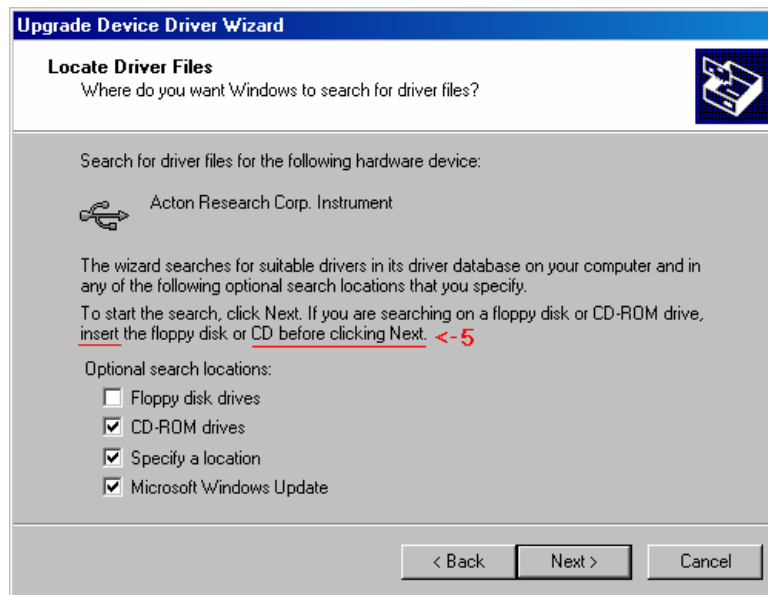


Figure 11. Locate Driver Files dialog





6. Select the CDROM drive (in this case, D:\ drive) and click on 'OK'.

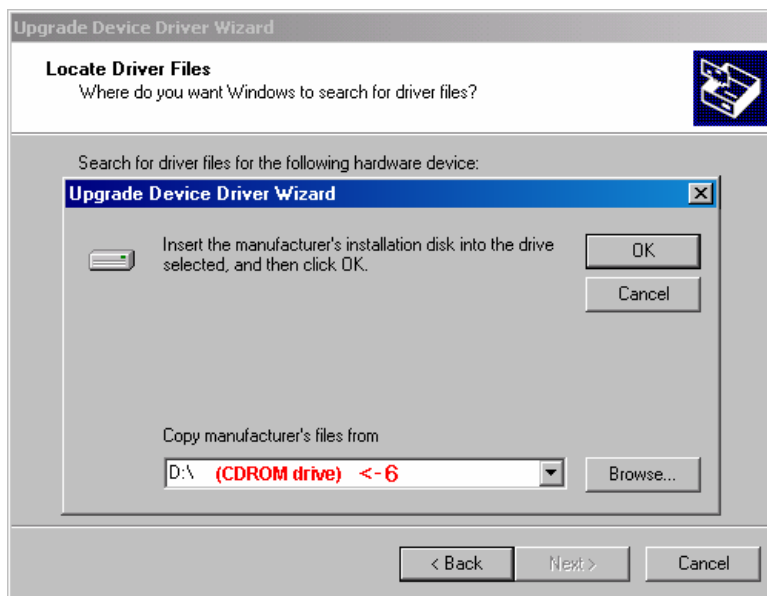


Figure 12. Location dialog

7. The wizard will search for the driver at the location you named. If it finds the driver on your hard drive as well, it will ask if you want to reinstall the current driver. Click on 'Next' to reinstall the driver.
8. When the 'Completing the Upgrade Device Driver Wizard' dialog appears, click on 'Finish' to close the wizard.
9. Close the Properties dialog, the Device Manager, and the Control Panel.
10. Now that the driver has been reloaded, we will bring up HyperTerminal to talk to the instrument (commands can be found in the manual). Go to page 4 for instructions on opening HyperTerminal.



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